Annals of Internal Medicine®

In the Clinic®

Community-Acquired Pneumonia

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Community-acquired pneumonia (CAP) can vary from a mild outpatient illness to a more severe disease requiring hospital admission and, at times, intensive care. CAP is the eighth leading cause of death, along with influenza, and is the leading cause of death from infectious diseases in persons in the United States older than 65 years. The key management decisions related to CAP are recognition and treatment in a timely and effective manner, defining the appropriate site of care (home, hospital, or intensive care unit [ICU]), and ensuring effective prevention. Health care–associated pneumonia (HCAP), or pneumonia in persons in contact with such health care environments as nursing homes and chronic hemodialysis centers or persons recently discharged from the hospital, can be caused by multidrug-resistant (MDR) organisms. Whether HCAP is best classified as a form of community-acquired or hospital-acquired pneumonia is controversial.

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Prevention

Who is at increased risk for CAP?

Persons with comorbid illness and the elderly are at increased risk for pneumonia and for having a more complex course. In 2006, there were nearly 1.3 million hospitalizations for pneumonia in the United States, and 57% were in persons older than 65 years (1). Comorbid illnesses that are associated with an increased incidence of CAP include respiratory disease, such as chronic obstructive pulmonary disease (COPD); cardiovascular disease; diabetes mellitus; and chronic liver disease. Other at-risk populations include those with HIV infection and other forms of immune suppression, as well as persons who have chronic kidney disease and those who have had splenectomy.

Cigarette smoking and alcohol abuse also often result in severe forms of CAP. Cigarette smoking is a risk factor for bacteremic pneumococcal infection (2). Other common illnesses in persons with CAP include cancer and any neurologic illness that predisposes to aspiration, including seizures. Immunization with pneumococcal vaccine in high-risk individuals and with influenza vaccine in at-risk patients during influenza season is beneficial for preventing CAP and many of its complications.

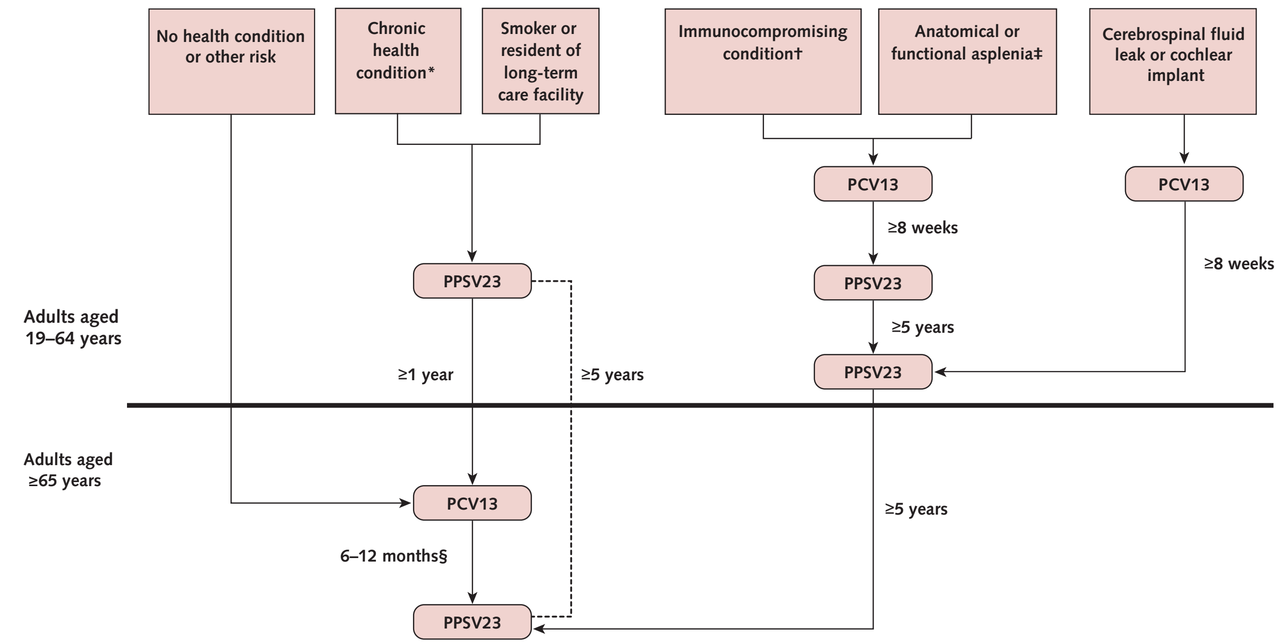
Who should receive pneumococcal vaccination and when should they receive it?

All individuals aged 65 years and older and other high-risk persons should receive pneumococcal vaccination. For persons without high-risk conditions, vaccination should be given at age 65 years. Patients with risk factors should be vaccinated when the risk is first identified, irrespective of age, with a special effort to review risk factors in persons older than 50 years. High-risk individuals who should be vaccinated include persons living in special environments (long-term care facilities), Alaskan natives or American Indians, and/or those who have any of the following conditions: chronic heart disease (congestive heart failure, cardiomyopathy but not hypertension), chronic lung disease (COPD but not asthma), diabetes mellitus, alcoholism, chronic liver disease, cigarette smoking, cerebrospinal fluid leaks, cochlear implants, functional or anatomical asplenia (including sickle cell disease), and immune-suppressing conditions (including HIV infection; congenital or acquired immune deficiencies; chronic renal failure; nephrotic syndrome; other immune suppression, including receipt of long-term corticosteroids; and cancer, including multiple myeloma, leukemia, lymphoma, and Hodgkin disease).

The 13-valent conjugate vaccine (PCV-13) and the 23-valent polysaccharide vaccine (PPS-23) are the 2 licensed adult pneumococcal vaccines available. All adults should receive PCV-13 in series with PPS-23. However, the timing of immunization varies by age and the presence of high-risk conditions. Because PCV-13 is more immunogenic, it should generally be given first, when vaccination is indicated, to all individuals who were not previously immunized, and followed 6–12 months later by PPS-23 (to add additional strain coverage). Immune-compromised patients younger than 65 years should receive PPS-23 only 8 weeks after PCV-13. Persons who have previously received PPS-23 should receive a dose of PCV-13 at least

1. American Lung Association. Trends in pneumonia and influenza and pneumonia morbidity and mortality; 2010.
2. Nuorti JP, Butler JC, Farley MM, et al. Cigarette smoking and invasive pneumococcal disease. Active Bacterial Core Surveillance Team. N Engl J Med. 2000;342:681-9. [PMID: 10706897]

Figure. **Recommended pneumococcal vaccine schedule and intervals, by age, health condition, and other risks.**



The dashed line represents the interval between the two PPSV23 doses. PCV13# = 13-valent pneumococcal conjugate vaccine; PPSV23# = 23-valent pneumococcal polysaccharide vaccine.

* Immunocompromising conditions are defined as congenital or acquired immunodeficiency (including B- or T-lymphocyte deficiency, complement deficiencies, and phagocytic disorders excluding chronic granulomatous disease), HIV infection, chronic renal failure, nephrotic syndrome, leukemia, lymphoma, Hodgkin disease, generalized malignancy, multiple myeloma, solid organ transplant, and iatrogenic immunosuppression (including long-term systemic corticosteroids and radiation therapy).

† Anatomical or functional asplenia is defined as sickle cell disease and other hemoglobinopathies, congenital or acquired asplenia, splenic dysfunction, and splenectomy.

‡ Chronic health conditions are defined as chronic heart disease (including congestive heart failure and cardiomyopathies, excluding hypertension), chronic lung disease (including chronic obstructive lung disease, emphysema, and asthma), chronic liver disease (including cirrhosis), chronic alcoholism, or diabetes mellitus.

§ Administer PPSV23 as soon as possible if the 6- to 12-month time window has passed. Reproduced from Reference 3.

1 year after the most recent vaccination. For those with immune-compromising conditions, a second dose of PPS-23 should be given 5 years after the first dose. For those who received either 1 or 2 doses of PPS-23 before 65 years of age, a repeated dose should be given at 65 years of age or older, provided that 5 years have passed since the prior dose. Although PCV-13 is generally given before PPS-23, current vaccine recommendations allow PPS-23 to be given first in persons younger than 65 years who have chronic health conditions but are not immune-suppressed and have not had splenectomy (3) (**Figure**).

Pneumococcal vaccination should be considered for anyone hospitalized for a medical illness. Although concerns have been raised about adverse reactions in patients who receive repeated vaccination, in 1 study fewer than 1% of patients who received at least 3 pneumococcal vaccinations had an adverse reaction, and no reaction was severe, even when vaccination was repeated in less than 6 years (4). Vaccination reduces the frequency of bacteremic pneumonia in healthy, immune-competent adults. Not all randomized, controlled trials (RCTs) have found reductions in the frequency of bacteremic pneumonia in adults with chronic illness, although case–control studies report reductions from 56%–81%.

In a double-blind RCT of 84 496 adults older than 65 years, PCV-13 prevented both bacteremic and nonbacteremic, vaccine strain–specific

3. Kim DK, Bridges CB, Harriman KH, et al. Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 and Older: United States, 2015. Ann Intern Med 2015;162:214-23. PMID: 25654609]
4. Walker FJ, Singleton RJ, Bulkow LR, et al. Reactions after 3 or more doses of pneumococcal polysaccharide vaccine in adults in Alaska. Clin Infect Dis. 2005;40:1730-5. [PMID: 15909258]

pneumococcal pneumonia and vaccine-type invasive pneumococcal pneumonia; however, it did not prevent all-cause CAP (5).

In an RCT of 1006 nursing home residents in Japan, PPS-23 significantly reduced the frequency and mortality of pneumococcal pneumonia, as well as the frequency– but not the mortality– of all-cause pneumonia (6).

The efficacy of pneumococcal vaccine has not been established in patients with sickle cell disease, chronic renal failure, immunoglobulin deficiency, Hodgkin disease, lymphoma, leukemia, or multiple myeloma. Use of the 7-valent conjugate vaccine in children led to herd immunity for older adults who live with immunized children, but necrotizing pneumonia caused by nonvaccine strains may be more common in children who receive this vaccine (7).

What is the role of influenza vaccination in the prevention of CAP and its complications?

All patients at increased risk for influenza complications and such persons as health care workers who are more likely to transmit the infection to high-risk patients (immune-suppressed, elderly, and those with chronic medical illness), should be immunized yearly. Adults aged 18–49 years can receive recombinant influenza vaccine, which does not contain the egg proteins (to which some patients believe themselves to be allergic) that are present in the inactivated vaccine or the live attenuated nasal vaccine. Live attenuated vaccine can be given intranasally to healthy, nonpregnant adults up to age 49 years. It should not be given to health care workers who are in contact with (and able to transmit the virus to) severely immune-compromised patients or to those with immunosuppression and chronic medical conditions. A high-dose influenza vaccine (4 times the standard dose) is available for individuals older than 65 years. In 1 randomized trial, this vaccine led to a higher antibody response and better protection against laboratory-confirmed influenza in persons older than 65 years than in those who received the standard influenza vaccine (8).

In a meta-analysis of 20 studies, influenza vaccine was shown to reduce pneumonia by 53%, hospitalization by 50%, and mortality by 68% (9). In addition, observational studies suggest that influenza vaccine can reduce all-cause mortality during influenza season by 27%–54% as well as being cost-effective due to its ability to reduce hospitalization rates for congestive heart failure and pneumonia in elderly persons (10). However, recent analyses question these benefits, noting that few RCTs have been conducted in these populations and that selection bias may lead to vaccination of healthier persons (11, 12).

Prevention... Persons at risk for CAP and its complications should be offered pneumococcal and influenza vaccination at the doses and frequencies described here. In nonvaccinated individuals in whom pneumococcal vaccine is indicated, give PCV-13 first followed by PPS-23 6–12 months later in immune-competent persons but after only 8 weeks in immune-suppressed patients. For persons who have previously received PPS-23, give 1 dose of PCV-13 at least 1 year after the prior immunization. Repeat PPS-23 vaccination after 5 years in persons older than 65 years who received the previous (1 or 2) doses before age 65 years, and after 5 years at any age in immune-suppressed patients. Influenza vaccine should be offered yearly to at-risk persons, including health care workers, with consideration of the high-dose vaccine in persons older than 65 years. Patients hospitalized with a medical illness should be considered for both of these vaccines.

CLINICAL BOTTOM LINE

5. Bonten MJM, Huijts SM, Bolkenbaas M, et al. Polysaccharide Conjugate Vaccine Against Pneumococcal Pneumonia in Adults. N Engl J Med. 2015;372:1114-1125. [PMID:25785969]
6. Maruyama T, Taguchi O, Niederman MS, et al. Efficacy of 23-valent pneumococcal vaccine in preventing pneumonia and improving survival in nursing home residents: double blind, randomised and placebo controlled trial. BMJ. 2010;340: c1004. [PMID: 20211953]
7. Bender JM, Ampofo K, Korgenski K, et al. Pneumococcal necrotizing pneumonia in Utah: does serotype matter? Clin Infect Dis. 2008;46:1346-52. [PMID: 18419434]
8. DiazGranados CA, Dunning AJ, Kimmel M, et al. Efficacy of high-dose versus standard-dose influenza vaccine in older adults. N Engl J Med. 2014;371:635-648. [PMID: 25119609]
9. Gross PA, Hermogenes AW, Sacks HS, et al. The efficacy of influenza vaccine in elderly persons. A meta-analysis and review of the literature. Ann Intern Med. 1995;123:518-27. [PMID: 7661497]
10. Nichol KL, Margolis KL, Wuorenma J, et al. The efficacy and cost effectiveness of vaccination against influenza among elderly persons living in the community. N Engl J Med. 1994;331:778-84. [PMID: 8065407]
11. Centers for Medicare & Medicaid Services. Hospital Quality Initiative Overview. Accessed at www.cms.hhs.gov/HospitalQualityInits/Downloads/Hospital-overview.pdf on 24 August 2009.
12. Nelson JC, Jackson ML, Weiss NS, Jackson LA. New strategies are needed to improve the accuracy of influenza vaccine effectiveness estimates among seniors. J Clin Epidemiol. 2009;62:687-94. [PMID: 19124221]

Which symptoms should lead clinicians to consider CAP?

Pneumonia usually presents with both respiratory and systemic symptoms, particularly in young patients and immune-competent persons. It should be suspected when the patient has cough, purulent sputum, pleuritic chest pain, dyspnea, chills, fever, night sweats, and weight loss. Fever and chills have a sensitivity of 50%–85% but may be absent in elderly persons. Dyspnea has a sensitivity of 70% for the diagnosis of CAP, whereas purulent sputum has a sensitivity of only 50%. Hemoptysis suggests necrotizing infection, such as lung abscess, tuberculosis, or gram-negative pneumonia. Many older patients and those with chronic illness have a weaker immune response, and the disease may go unrecognized because the patient has only nonrespiratory symptoms. These symptoms include confusion, weakness, lethargy, falling, poor oral intake, and decompensation of a chronic illness (for example, congestive heart failure). Most patients with CAP present with an acute illness 1–2 days in duration, but symptoms may be present for longer periods in elderly persons.

Which organisms cause CAP?

The most commonly identified bacterial pathogens for CAP are *Streptococcus pneumoniae* (pneumococcus); *Haemophilus influenzae;* and atypical pathogens, such as *Mycoplasma pneumoniae, Chlamydophila pneumoniae*, and *Legionella*. Patients older than 65 years and those with alcoholism, noninvasive disease, antibiotic therapy within 3 months, multiple medical comorbid conditions, exposure to children in a day care center, or immunosuppressive illness are more likely to have drug-resistant pneumococcus (DRSP) than other patients (**Table**).

A cohort study of 3339 patients with invasive pneumococcal infection found that if the patient had received penicillin, macrolide, fluoroquinolone, or trimethoprim–sulfamethoxazole therapy 3 months before the onset of bacteremia, the organism was more likely to be resistant to the antibiotic recently received (13).

Viruses can also cause CAP, and 1 recent study found viruses in 18% of all patients who had paired serologies. The most com-

Table. **Modifying Factors That Increase the Risk for Infection With Specific Pathogens**

Organism	Risk Factor
Penicillin-resistant and drug-resistant pneumococci	Age >65 years β-lactam therapy within the past 3 months Alcoholism Immune-suppressive illness (including therapy with corticosteroids) Multiple medical comorbid conditions Exposure to a child in a day care center
Enteric gram-negative bacteria	Residence in a nursing home Underlying cardiopulmonary disease Multiple medical comorbid conditions Recent antibiotic therapy
Pseudomonas aeruginosa	Structural lung disease (bronchiectasis) Corticosteroid therapy (prednisone, >10 mg/d) Broad-spectrum antibiotic therapy for >7 d in the past month Malnutrition

13. Toronto Invasive Bacterial Disease Network. Predicting antimicrobial resistance in invasive pneumococcal infections. Clin Infect Dis. 2005;40: 1288-97. [PMID: 15825031]

mon viral organisms were influenza and parainfluenza virus, followed by respiratory syncytial virus and adenovirus. Viruses are also common in patients with severe CAP, especially with the advent of molecular diagnostic methods. In a study of 198 ICU patients with CAP or HCAP, viruses were present in 72, using real-time polymerase chain reaction (PCR) testing or shell vial cultures of either bronchoalveolar lavage or nasal swabs. Rhinovirus, parainfluenza, influenza, and respiratory syncytial virus were most common (14). In another study of 468 ICU patients, viruses were present in 23% of adults, using PCR testing of nasopharyngeal swabs (15). Gram-negative bacteria have been found in up to 10% of patients with CAP (although some of these patients are now classified as having HCAP), particularly those with a history of chronic cardiopulmonary disease, residence in a nursing home, multiple medical comorbid conditions, recent antibiotic therapy, renal insufficiency, chronic liver disease, diabetes, or active cancer (16). *Pseudomonas aeruginosa* should be considered in persons with bronchiectasis, recent hospitalization, or recent antibiotic therapy. Although anaerobic organisms should be considered when aspiration is a possibility (for example, in elderly patients with neurologic or swallowing disorders), in 1 study the most common organisms identified in persons at risk for aspiration were gram-negative bacteria (17). *Klebsiella pneumoniae* has been reported in patients with alcoholism. Although some studies suggest that HCAP pathogens are more similar to those in hospital-acquired pneumonia than to those in CAP, not all studies confirm these findings. Patients with HCAP who are most at risk for drug-resistant organisms are those with poor functional status, severe illness, recent antibiotic therapy, and recent hospitalization (18). Methicillin-resistant *S. aureus* (MRSA) can occur in patients with HCAP and in previously healthy persons after influenza, particularly in the form of a severe, bilateral, necrotizing infection (19, 20).

What is the role of history and physical examination in the diagnosis of CAP?

History and physical examination are valuable for suggesting the presence of pneumonia, predicting the cause, and helping to define severity. The history can identify risk factors for HCAP, such as hospitalization or antibiotic therapy in the past 90 days, residence in a long-term care facility, long-term dialysis, outpatient wound care, or home infusion therapy. It can also identify patients who might have a less common cause of pneumonia, such as bird (*Chlamydia psittaci, Cryptococcus neoformans*) or bat (*Histoplasma capsulatum*) exposure or travel to the southwestern United States (endemic fungi, such as coccidioidomycosis). In addition to fever, dyspnea, cough, and pleuritic chest pain, symptoms can include confusion, weakness, falling, lethargy, reduced oral intake, or deterioration due to a chronic illness (such as congestive heart failure).

Clinical examination findings suggestive of pneumonia include fever or hypothermia, tachypnea, crackles, bronchial breath sounds on auscultation, and pleural effusion. Specific findings that are associated with a poor outcome include a respiratory rate greater than 30 breaths/min, diastolic blood pressure less than 60 mm Hg, systolic blood pressure less than 90 mm Hg, heart rate greater than 125 beats/min, and temperature less than 35 °C or greater than 40 °C. An elderly

14. Choi SH, Hong SB, Ko GB, et al. Viral infection in patients with severe pneumonia requiring intensive care unit admission. Am J Respir Crit Care Med. 2012;186: 325-32. [PMID: 22700859]
15. Wiemken T, Peyrani P, Bryant K, et al. Incidence of respiratory viruses in patients with community-acquired pneumonia admitted to the intensive care unit: results from the Severe Influenza Pneumonia Surveillance (SIPS) project. Eur J Clin Microbiol Infect Dis. 2013;32:705-10. [PMID: 23274861]
16. Arancibia F, Bauer TT, Ewig S, et al. Community-acquired pneumonia due to gram-negative bacteria and pseudomonas aeruginosa: incidence, risk, and prognosis. Arch Intern Med. 2002;162:1849-58. [PMID: 12196083]
17. El-Solh AA, Pietrantoni C, Bhat A, et al. Microbiology of severe aspiration pneumonia in institutionalized elderly. Am J Respir Crit Care Med. 2003;167:1650-4. [PMID: 12689848]
18. Maruyama T, Fujisawa T, Okuno M, et al. A new strategy for health care-associated pneumonia: a 2-year prospective multicenter cohort study using risk factors for multidrug-resistant pathogens to select initial empiric therapy. Clin Infect Dis. 2013;57:373-83. [PMID: 23999080]
19. Carratalà J, Mykietiuk A, Fernández-Sabé N, et al. Health care-associated pneumonia requiring hospital admission: epidemiology, antibiotic therapy, and clinical outcomes. Arch Intern Med. 2007;167:1393-9. [PMID: 17620533]
20. Infectious Diseases Society of America. Infectious Diseases Society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. Clin Infect Dis. 2007;44 Suppl 2:S27-72. [PMID: 17278083]

person with CAP may present with nonrespiratory symptoms and findings because of an impaired immune and inflammatory response.

When should clinicians use chest radiography?

A chest radiograph should be obtained in all patients with clinical features suggesting CAP to define the presence of parenchymal lung infection and to identify certain pneumonia complications. Clinical diagnosis of pneumonia is often inaccurate and has an overall sensitivity ranging from 70%–90% and a specificity ranging from 40%–70% (20, 21). Elderly and immunosuppressed patients can have radiographic evidence of pneumonia without clear-cut clinical features. In 1 study, measurement of serum procalcitonin levels added to physical examination data increased the predictive value of an abnormal chest radiograph (22).

When history (cough, fever, dyspnea, pleuritic pain) and physical examination findings (focal crackles, temperature ≥38 °C) were used to predict the presence of radiographic pneumonia in a study of 129 patients with lower respiratory tract infection (26 with pneumonia), no combination of findings was highly accurate. The positive predictive value of each finding varied from 17%–43% (21).

It is especially important to have a chest radiograph if the diagnosis is questionable or if pleural effusion, lung abscess, necrotizing pneumonia, or multilobar illness is suspected. Radiographs can specifically aid patient management if findings of severe illness are present (bilateral, multilobar, or rapidly expanding infiltrates), but patterns only rarely suggest a specific diagnosis or cause (e.g., tuberculosis, *P. jiroveci*). If pleural effusion is present, confirmation should be obtained with a decubitus film, thoracic ultrasonography, or computed tomography. Pneumonia should be assumed to be present even in the absence of a radiographic infiltrate if the patient has a convincing history and focal physical findings; a follow-up radiograph may show an infiltrate. Interobserver variability in chest radiographic interpretation was shown in 1 study that compared the readings of at least 2 radiologists. Positive agreement (59%) was less frequent than negative agreement (94%) (23).

What is the role of other laboratory tests?

For outpatients, perform pulse oximetry to assess oxygenation. No other tests are needed. For inpatients, additional testing is done to define disease severity and identify the cause. Measure pulse oximetry in all patients and arterial blood gases in those suspected of having carbon dioxide retention. Even with extensive diagnostic testing, a specific cause is found in fewer than half of all patients. Sputum should be collected for Gram stain and culture before therapy is started in patients suspected of being infected with a drug-resistant or unusual pathogen, but evaluation of sputum should be done only if it is of good quality and processed rapidly. Rapid diagnostic testing of respiratory secretions with molecular methods (such as PCR) is now becoming widely available, although their value for patient management is uncertain because they may be too sensitive and unable to distinguish colonization from infection. In patients with severe pneumonia, 2 sets of blood cultures should be collected and urine tested for *Legionella* and pneumococcal antigens. Limit blood cultures to patients with severe illness; these cultures are positive in only 10%–20% of all patients with CAP. In low-risk patients (without severe illness), the incidence of false-positive results may exceed the incidence of true-positive re-

21. Graffelman AW, le Cessie S, Knuistingh Neven A, et al. Can history and exam alone reliably predict pneumonia? J Fam Pract. 2007;56:465-70. [PMID: 17543257]
22. Müller B, Harbarth S, Stolz D, et al. Diagnostic and prognostic accuracy of clinical and laboratory parameters in community-acquired pneumonia. BMC Infect Dis. 2007;7:10. [PMID: 17335562]
23. Hopstaken RM, Witbraad T, van Engelshoven JM, Dinant GJ. Inter-observer variation in the interpretation of chest radiographs for pneumonia in community-acquired lower respiratory tract infections. Clin Radiol. 2004;59:743-52. [PMID: 15262550]

sults (20, 24). Culture an endotracheal aspirate in patients who are intubated and mechanically ventilated.

A study of 13 043 Medicare patients identified the following predictors of true-positive results on blood culture: no previous receipt of antibiotics, underlying liver disease, systolic blood pressure <90 mm Hg, temperature <35 °C or >40 °C, pulse >125/min, blood urea nitrogen levels >10.71 mmol/L (30 mg/dL), serum sodium levels <130 mmol, and leukocyte count <5 or >20 × 10^9 cells/L. The diagnostic yield of blood cultures increased in patients with 1 or more of the above predictors and in those who had not received antibiotics before blood collection (24).

Do not perform serologic tests for viruses and atypical pathogens because they require convalescent titers 6–8 weeks after the initial test to identify infection. Rapid diagnostic tests that can evaluate for viral pathogens on nasopharyngeal swabs, such as rapid antigen testing, direct fluorescent antibody, or PCR, are available. However, the role of these tests in managing patients with CAP and in guiding antibiotic selection are not yet established, although some studies have shown that they may reduce unnecessary antibiotic use and increase antiviral use, especially when results are positive for influenza. Regardless, establishing a specific cause is usually not necessary because empirical therapy is effective. For example, when pathogen-directed treatment was compared with empirical treatment using a broad-spectrum antibiotic, the 2 groups did not significantly differ in the length of hospital stay, 30-day mortality, clinical failure, or resolution of fever (25).

Measurement of serum levels of C-reactive protein or procalcitonin may be helpful to define which patients have bacterial pneumonia, to decide whether to start antibiotic therapy, to determine prognosis, and to provide supportive information in the site-of-care decision, but their role in the routine management of CAP is not defined. Serum levels of procalcitonin identify patients who can benefit from antibiotic therapy; levels are elevated with bacterial and *Legionella* infection, but not always with other atypical pathogen infection, and not with viral infection. Serial measurements of procalcitonin levels may help guide when to stop antibiotic therapy (26).

A randomized trial of 302 patients with CAP compared patients managed by usual care with those managed by an algorithm recommending antibiotics and the duration of therapy. The algorithm was based on serial measurement of procalcitonin levels using the highly sensitive Kryptor assay. Procalcitonin levels were measured on admission and after 6–24 hours, 4 days, 6 days, and 8 days. The procalcitonin-guided group had significantly fewer antibiotic prescriptions on admission and less antibiotic use, and the duration of therapy was reduced from 12 to 5 days with similar clinical success (27). Measurement of serum procalcitonin, using the Kryptor assay at the time of admission, has been shown in 4 randomized studies (of both single-center and multicenter design) to reduce antibiotic use. In these studies, antibiotic therapy was discouraged when the procalcitonin level was <0.25 µg/L, with no adverse effects on mortality (26).

What other disorders should clinicians consider in patients suspected of having CAP?

If the patient does not respond to empirical therapy after 48–72 hours, consider the possibility of a virus or an unusual bacterial pathogens, such as *Mycobacterium tuberculosis* (which may be masked by a partial response to empirical quinolone therapy of CAP) or endemic fungi (histoplasmosis, coccidioidomycosis, blastomycosis). Also consider noninfectious possibilities, such as bronchiolitis obliterans with organizing pneumonia, pulmonary vasculitis, hypersensitivity pneumonitis, interstitial diseases, lung cancer, lymphangitic carcinoma, bronchoalveolar cell carcinoma, lymphoma, and congestive heart failure. If the patient's condition de-

24. Metersky ML, Ma A, Bratzler DW, Houck PM. Predicting bacteremia in patients with community-acquired pneumonia. Am J Respir Crit Care Med. 2004; 169:342-7. [PMID: 14630621]
25. van der Eerden MM, Vlaspolder F, de Graaff CS, et al. Comparison between pathogen directed antibiotic treatment and empirical broad spectrum antibiotic treatment in patients with community acquired pneumonia: a prospective randomised study. Thorax. 2005;60: 672-8. [PMID: 16061709]
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27. Christ-Crain M, Stolz D, Bingisser R, et al. Procalcitonin guidance of antibiotic therapy in community-acquired pneumonia: a randomized trial. Am J Respir Crit Care Med. 2006; 174:84-93. [PMID: 16603606]

teriorates after an initial response to therapy, consider pulmonary embolus; antibiotic-induced colitis; and the pneumonia complications of empyema, meningitis, and endocarditis.

When should clinicians consider specialty consultation for diagnosis, and which types of specialists should they consult?

Consultation is most valuable to clarify diagnostic questions when patients do not respond to initial empirical therapy. An infectious disease specialist can help identify infectious complications of pneumonia and unusual infections, such as *Coxiella burnetii* (Q fever), *Burkholderia pseudomallei* (melioidosis), *Mycobacterium tuberculosis* (which may be masked by a partial response to empirical quinolone therapy of CAP) , endemic fungi (histoplasmosis, coccidioidomycosis, blastomycosis), *Pasteurella multocida, Bacillus anthracis, Actinomyces Israeli, Francisella tularensis* (tularemia), *Yersinia pestis* (plague), or *Chlamydia psittaci* (psittacosis). A pulmonary specialist can help identify inflammatory lung disease and pulmonary embolus and perform bronchoscopy and transbronchial biopsy. A surgeon can perform thoracoscopic or open lung biopsy.

Diagnosis... History is particularly valuable for defining risk factors for specific pathogens, and physical findings help define disease severity. Clinical findings are less dramatic in elderly persons. Confirm the diagnosis of CAP with a chest radiograph, although this test is not always definitive early in the course of illness. Laboratory testing has limited value; however, it can be used in conjunction with other data to define severity and to identify systemic and respiratory complications. Diagnosing specific pathogens early is less useful because most initial therapy is empirical. If the patient does not respond to initial therapy, consult specialists and consider bronchoscopy and lung biopsy for a definitive diagnosis.

CLINICAL BOTTOM LINE

Treatment

How should clinicians determine whether a patient with CAP requires outpatient, inpatient, or ICU care?

Many site-of-care decisions can be facilitated with the pneumonia severity index (PSI) or the British Thoracic Society (BTS) rule. These tools predict risk for death. Patients at high risk are generally managed in the hospital, and those at highest risk are managed in the ICU. The PSI stratifies patients into 5 categories by using a scoring system based on patient age, comorbid illness, physical examination findings, and laboratory data. In general, patients in classes I and II are treated as outpatients, those in class III have the site-of-care decision based on careful clinical assessment, and those in classes IV and V are generally admitted to the hospital. The BTS rule has been condensed into the "CURB-65," which is based on the presence of **C**onfusion, blood **U**rea nitrogen >7.0 mmol/L (19.6 mg/dL), **R**espiratory rate ≥30 breaths/min, systolic **B**lood pressure <90 mm Hg or diastolic blood pressure <60 mm Hg, and age ≥65 years. Patients with at least 2 of these criteria are usually admitted to the hospital, whereas those with at least 3 criteria are considered for ICU admission (20, 28, 29).

One prospective study of 3181 patients seen in 32 emergency departments compared the PSI with the CURB and CURB-65 criteria and found that both approaches accurately identified low-risk patients. CURB-65 was better for predicting mortality in high-risk patients (28). In another prospective study of 1651 patients, measurement of serum procalcitonin levels supplemented data obtained by prognostic scoring, and patients who had low levels of procalcitonin had low mortality, regardless of PSI class or number of CURB-65 points (29).

Current guidelines support ICU care if the patient needs assisted ventilation, has septic shock requiring vasopressors, or has at least 3 of the following: respiratory rate ≥30 breaths/min, PaO_2/FiO_2 ratio ≤250, multilobar infiltrates, confusion or disorientation, blood urea nitrogen ≥7.1 mmol/L (20 mg/dL), leukocyte count $<4 \times 10^9$ cells/L, platelet count $<100 \times 10^9$ cells/L, temperature lower than 36 °C, and hypotension requiring aggressive fluid resuscitation (20). One study found that leukopenia, thrombocytopenia, and hypothermia were each present in <5% of all patients; further, omitting these criteria and replacing them with acidosis simplified the prediction of need for ICU admission without a loss of accuracy (30).

What is the role of nondrug therapies?

In outpatients, nondrug therapy should focus on encouraging oral hydration. For hospitalized patients, nondrug therapies include intravenous hydration and oxygen for hypoxemia. Chest physiotherapy has not been widely studied, but it has been shown to improve the outcome of patients with pneumonia who have more than 30 mL/d of sputum and impaired clearance of secretions (31). A meta-analysis of 6 randomized trials in 434 patients evaluated the following 4 types of chest physiotherapy: conventional chest physiotherapy, active cycle breathing, osteopathic manipulation, and positive expiratory pressure. No method reduced mortality, whereas osteopathic manipulation and positive expiratory pressure reduced the duration of hospital stay by 2.02 and 1.4 days, respectively (32). In the severely ill ICU patient, nondrug therapy can include noninvasive ventilatory support and mechanical ventilation for those with respiratory failure.

Which antibiotics should be prescribed for outpatients?

For patients with no cardiopulmonary disease and no factors that increase risk for infection with DRSP or enteric gram-negative bacteria (**Table**), prescribe a macrolide (azithromycin, clarithromycin, or erythromycin) or doxycycline (**Appendix Table**, available at www.annals.org). For outpatients who have cardiopulmonary disease or factors that increase the risk for infection with DRSP or enteric gram-negative bacteria, prescribe an antipneumococcal quinolone (levofloxacin or moxifloxacin) or a combination of a β-lactam (amoxicillin, 3 g/d; amoxicillin–clavulanate, cefpodoxime, or cefuroxime) with a macrolide or doxycycline. If the patient has received an antibiotic in the past 3 months, avoid using an antibiotic of the same class.

How long should outpatients continue antibiotic treatment?

The duration of therapy should be based on the patient's clinical response, severity of illness, and probable pathogen. Outpatients with mild-to-moderate CAP can be treated for as few as 5 days if clinical response is good, there has been no fever for 48 to 72 hours, and there are no signs of extrapulmonary infection. Azithromycin has such a long half-life that therapy for 1 or 3 days may be effective.

A meta-analysis of 15 RCT studying mild-to-moderate CAP found that therapy for 7 days or fewer was as effective as longer therapy with regard to clinical failure, mortality, adverse events, and bacteriologic eradication. The trials compared only monotherapies, and 13 of the short-duration trials used azithromycin, fluoroquinolones, or ketolides, which provide coverage for both pneumococcus and atypical pathogens. Only 2 short-duration trials used β-lactam monotherapy (33).

How should clinicians follow patients during outpatient treatment?

Up to 10% of patients initially managed at home do not respond to outpatient therapy and require hospitalization. To identify these patients early, the physician and patient should agree on a plan to monitor the response to therapy. Ask patients to measure their temperature orally every 8 hours and to report if it exceeds 38.3 °C (101 °F) or does not decrease below 37.2 °C (99 °F) after 48 hours. Encourage patients to drink at least 1 to 2 quarts of liquid daily and to report if they cannot achieve this goal. Instruct patients to report symptoms of chest pain, severe or increasing shortness of breath, or lethargy. Encourage patients to take their antibiotics on schedule

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and to continue taking them even after they begin feeling better and until they have taken all of them.

If the response to therapy is satisfactory, ask the patient to return for an examination in 10–14 days. Give pneumococcal and influenza vaccinations at the outpatient visit if they have not previously been given. Obtain a repeated chest radiograph no sooner than 1 month after starting pneumonia therapy to screen for nonresolution of infiltrates, which could be due to lung cancer, inflammatory disease, or an unusual infection (20). Assess the patient's overall health and look for decompensated comorbid illness to attempt to prevent future episodes of pneumonia.

When patients require hospitalization, how soon after admission should antibiotics be started, and which antibiotics should patients receive if they do not need ICU care?

Patients should receive initial antibiotic therapy as soon as possible after the diagnosis is established and before they leave the emergency department. Although therapy within 4 hours of arrival to the hospital has been associated with reduced mortality in some studies, undue emphasis on early therapy can lead to unnecessary use of antibiotics and associated complications (11, 34). In 1 study, the final diagnosis of pneumonia in patients suspected of having pneumonia in the emergency department decreased from 75.9% to 58.9% after the initiation of a program to give more patients antibiotics within 4 hours of arrival in the emergency department (35).

Although guidelines recommend that hospitalized patients who are not in the ICU receive intravenous azithromycin if they have no cardiopulmonary disease and no factors that increase the risk for DRSP or gram-negative bacteria (20, 36), very few such patients are hospitalized, and this is not a commonly used approach. Most hospitalized patients not in the ICU have cardiopulmonary disease or factors that increase the risk for DRSP or gram-negative bacteria, and they should receive an intravenous or oral quinolone (levofloxacin, 750 mg/d, when renal function is normal, or moxifloxacin, 400 mg/d) or the combination of a β-lactam (cefotaxime, ceftriaxone, ampicillin–sulbactam, or high-dose ampicillin, but not cefuroxime) with a macrolide or doxycycline (20). The addition of a macrolide to a β-lactam has been associated with reduced mortality and hospital stay (37), even for bacteremic pneumococcal pneumonia (37). In contrast, in a randomized trial of 580 immune-competent patients, β-lactam monotherapy was not inferior to a β-lactam–macrolide combination in patients with moderately severe CAP. However, patients with atypical pathogens and those in PSI class IV had delayed clinical stability with β-lactam monotherapy when compared with those receiving combination therapy (38). In another trial of admitted patients with nonsevere CAP, β-lactam monotherapy was equivalent to a β-lactam–macrolide combination for 30 day mortality (which was <10%) (39). Specific β-lactams, such as ceftriaxone and cefotaxime, are preferred if DRSP is suspected because they are effective at mean inhibitory concentrations up to 2 mg/L (40). However, 1 study showed increased mortality when cefuroxime was used in patients with bacteremic DRSP (41).

One international study of 4337 hospitalized patients with CAP showed that approximately 20% had evidence of atypical pathogen infection and that therapy directed against these organisms decreased the time to clinical stability, length of stay, and both total and CAP-

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related mortality (42). However, another study of 2209 hospitalized Medicare patients with bacteremic pneumonia found that therapy directed at atypical pathogens reduced 30-day mortality and 30-day readmission rate, but the benefits occurred only with macrolides and not with fluoroquinolones (43).

For patients with HCAP admitted to the hospital but not the ICU, antibiotic choice should be individualized based on risk factors for MDR pathogens. A CAP regimen is effective for most patients, but therapy directed at multidrug-resistant, gram-negative bacteria and MRSA may be needed for patients with nonsevere pneumonia and at least 2 MDR risk factors (recent hospitalization, recent antibiotics, poor functional status, immune suppression) or severe pneumonia with at least 1 MDR risk factor. Therapy for HCAP in patients at risk for MDR pathogens usually requires dual pseudomonal therapy plus therapy directed at MRSA.

A prospective study of 321 HCAP patients showed that with the use of an algorithm, 93% received appropriate therapy, but this was achieved with only 53% requiring broad-spectrum empirical therapy (18). The algorithm stratified patients on the basis of severity of illness (admission to the ICU or not) and the presence of other risk factors for MDR pathogens (recent antibiotic therapy, recent hospitalization, poor functional status, and immune suppression). Patients with nonsevere illness and at least 2 MDR risk factors, or those with severe illness and at least 1 MDR risk factor, were treated empirically with broad-spectrum, multidrug therapy, whereas all others received CAP regimens. Through use of this approach, mortality was 13.7%.

Which antibiotics should be given to patients admitted to the ICU?

No patient in the ICU should receive empirical monotherapy. Assess these patients for risk factors for *P. aeruginosa*. Treat those without risk factors with intravenous ceftriaxone or cefotaxime plus either azithromycin or a quinolone, such as levofloxacin or moxifloxacin. Treat patients who have risk factors with an intravenous, antipseudomonal β-lactam (cefepime, piperacillin–tazobactam, imipenem, meropenem) plus an intravenous quinolone effective against *P. aeruginosa* (ciprofloxacin or high-dose levofloxacin). Alternatively, treat patients with risk factors with an intravenous, antipseudomonal β-lactam combined with an aminoglycoside (amikacin, gentamicin, or tobramycin) plus either an intravenous macrolide (azithromycin or erythromycin) or intravenous antipneumococcal quinolone (levofloxacin or moxifloxacin). In studies of patients admitted to the ICU with severe CAP, mortality was reduced when combination therapy was used; monotherapy, even with a quinolone, was not as effective.

A meta-analysis of 28 observational studies, involving nearly 10 000 critically ill patients found that macrolide use (generally in a combination regimen) was associated with an 18% reduction in mortality compared with nonmacrolide regimens and that a β-lactam–macrolide combination had a trend toward reduced mortality compared with a β-lactam–quinolone regimen (44). In patients with bacteremic pneumococcal pneumonia and critical illness, studies have found that mortality was lower with combination therapy than with monotherapy (45).

If community-acquired MRSA is suspected, add either linezolid alone or consider vancomycin combined with clindamycin, because both of these regimens are antibacterial and inhibit production of bacterial toxins. Vancomycin alone is antibacterial but cannot inhibit toxin production (46). These regimens are recommended, even though the organisms are often sensitive in vitro to trimethoprim–sulfamethoxazole and quinolones.

What are the other components of ICU care for CAP?

Hydration should be ensured and supplemental oxygen and chest physiotherapy should be considered. However, the main determination is whether ventilatory support for respiratory failure is needed. Intubation and mechanical ventilation are required in patients who have oxygen saturation less than 90% on maximal mask oxygen, inability to clear secretions, inability to protect

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the airway, or hypercarbia. If the patient has only hypoxemia or hypercarbia and is alert and cooperative, it may be possible to use noninvasive positive pressure ventilation. This therapy may be associated with fewer complications than endotracheal intubation—including ventilator-associated pneumonia—but if it is used, the clinician must ensure that the patient can expectorate any respiratory secretions. Consider systemic corticosteroids, especially if relative adrenal insufficiency is suspected, or if a patient with pneumococcal pneumonia has associated meningitis. Routine use of systemic corticosteroids in severe CAP is not recommended, but patients with severe illness and high levels of systemic inflammation may benefit from adjunctive systemic corticosteroids. Because many patients with severe CAP also have systemic sepsis, consider aggressive hydration, vasopressors, and measurement of serum lactate.

In 1 study of 40 patients with severe CAP when random serum cortisol levels were measured in the first 72 hours, 65% of patients met criteria for adrenal insufficiency and 63% of the 19 patients with CAP and septic shock also had adrenal insufficiency (47). In 4 studies, including randomized trials, evidence was inconsistent for a benefit from routine corticosteroid therapy; however, if the patient required corticosteroids for another reason (such as underlying COPD), they seemed to cause no harm (48). A multicenter RCT of severe CAP compared 61 patients given 0.5 mg/kg methylprednisolone every 12 hours for 5 days with 59 patients given placebo. All patients had severe CAP and a C-reactive protein level greater than 150 mg/L on admission. The group treated with corticosteroids had less treatment failure, with no difference in mortality (49).

When can clinicians switch hospitalized patients from intravenous to oral antibiotics?

A switch from intravenous to oral antibiotics is indicated once the symptoms of cough, sputum production, and dyspnea improve; the patient is afebrile on 2 occasions 8 hours apart; and he or she is able to receive oral medications. This switch can be made as early as 24–48 hours after admission and is made by day 3 in up to half of all patients. Changing to oral therapy can be done safely even if pneumococcal bacteremia has been documented, although these patients may take longer to respond. Longer durations of therapy are usually needed for patients infected with *P. aeruginosa* or *S. aureus* and for those with extrapulmonary complications, such as empyema or meningitis, but should be individualized to specific patient situations. Select an oral regimen that covers all organisms isolated in blood or sputum cultures and reflects the intravenous therapy. For some patients, this means a β-lactam–macrolide combination or quinolone monotherapy. Patients who have responded to a β-lactam–macrolide combination can be continued on macrolide monotherapy unless cultures justify dual therapy.

To facilitate a switch to oral therapy, hospitals should consider using a standing order set supplemented by prospective case management. In a cohort study, each patient was managed in 3 successive periods with conventional therapy, a guideline-based order set supported by prospective case management that provided feedback to clinicians, and a guideline-based order set alone. Time to clinical stability was similar in all 3 periods, but prospective case management led to the greatest reductions in the time to oral antibiotics, time from oral therapy to discharge, and overall length of stay (50).

When should a consultation be requested for hospital patients, and which types of specialists or subspecialists should be consulted?

An infectious disease or pulmonary consultation is appropriate if there are questions about the selection of initial antibiotic therapy or when the patient does not respond to initial therapy. A pulmonary or critical care physician should be consulted for patients with severe illness to decide about using vasopressors, determine the appropriate site of care, decide about the need for ventilatory support, and to aid in managing the mechanical ventilator. Pulmonary consultation should be requested if pleural effusion is documented and help is needed with a decision regarding thoracentesis. Pulmonary or thoracic surgical consultation is appropriate for placement of a chest tube if a complicated parapneumonic effusion or empyema is found on thoracentesis, because early therapy can reduce hospital stay and avoid complications. A thoracic surgeon can also perform surgical decortication for advanced and loculated pleural effusion and empyema. Cardiology consultation may be needed if complications of cardiac ischemia arise or in cases of congestive heart failure. In a study of 170 patients with pneumococcal pneumonia, 19.4% had at least 1 major cardiac event, including 12 with acute myocardial infarction, 8 with new-onset atrial fibrillation or ventricular tachycardia, and 13 with newly diagnosed or worsening heart failure without other cardiac complications. Patients with cardiac events had a significantly higher mortality rate (27.3% vs. 8.8%) (51).

When can inpatients be discharged from the hospital?

Patients can be discharged once a switch to oral therapy is made and coexisting medical conditions are under control. No benefit for continued hospital observation has been proven. In 1 study, two thirds of clinically stable patients were observed while receiving oral therapy before discharge, and no deterioration occurred during this period (52). Another study compared patients who remained in the hospital for 1 day after the switch from intravenous to oral therapy with those discharged on the day of the switch. The study excluded patients with complicated pneumonia, those who were not eligible to be switched to oral therapy, and those with lengths of stay less than 3 days or more than 7 days. There were no differences in mortality or 14-day readmission rate (53). Therapy may need to be continued after discharge, but the total duration of therapy is usually 5–7 days.

What are the indications for follow-up chest radiography?

Patients hospitalized for pneumonia do not need a routine chest radiograph before discharge, but those who do not reach clinical stability and those who deteriorate despite therapy require an aggressive evaluation, including an early follow-up chest radiograph. If the patient has a good clinical response to therapy, a chest radiograph should not be repeated any earlier than 4–6 weeks after initial therapy. Radiographic resolution usually lags behind clinical resolution by about 6–8 weeks, but early improvement is usually substantial. In a prospective study of patients aged 70 years or older, 58% had a clear chest radiograph after 3 weeks, but it took 12 weeks until at least 75% had radiographic resolution of CAP. Predictors of slow resolution included a high comorbidity index, bacteremia, multilobar involvement, and infection with enteric gram-negative bacteria (54).

How can patients prevent recurrent CAP?

Patients with CAP should be up-to-date with pneumococcal and influenza vaccinations; avoid smoking cigarettes; and receive optimal therapy for comorbid illnesses, such as congestive heart failure and COPD. In addition, careful evaluation is needed for medical conditions that could predispose to recurrent infection. One study found new comorbid conditions in 6% of patients with CAP, including diabetes mellitus, cancer, COPD, and HIV infection (55). The patient should be evaluated for aspiration risk factors. If pneumonia recurs in the same location, consideration should be given to the possibility of bronchiectasis, aspirated foreign body, or endobronchial obstruction. If the patient has recurrent pneumonia or pneumonia with an unusual pathogen, immune deficiency may be present.

The 30-day readmission rate for CAP patients varied from 16.8%–20.1% in a review of 12 studies (56). Pneumonia itself was the cause of readmission in only 17.9%–29.4% of patients; however, other common causes were exacerbations of congestive heart failure or COPD. Patients with HCAP have a higher risk for readmission than patients with CAP (57).

Treatment... The most important clinical decisions in the treatment of CAP include determining the site of care (outpatient, hospital, or ICU), selecting antibiotic therapy, delivering supportive care (oxygen, hydration), and determining the need for ventilatory support. Antibiotic therapy differs for outpatients, inpatients, and those in the ICU. However, all patients should receive timely empirical therapy directed at pneumococcus, atypical pathogens, and other organisms when they have risk factors for those organisms. The PSI and the CURB-65 aid decisions about the site of care. Patients should be managed in the ICU if they require ventilatory or vasopressor support or close observation. Consultation should take place in cases of severe disease and when patients do not respond to initial therapy or have complications. Inpatients can be transitioned to oral antibiotics after treatment response occurs and they are clinically stable, after which the patient can be discharged and managed on an outpatient basis. Routine follow-up chest radiography should be delayed for 4–6 weeks if the patient is responding well to therapy. During follow-up, patients should be monitored for undiagnosed or ineffectively managed comorbid illness, up-to-date on pneumococcal and influenza vaccinations, and encouraged to avoid cigarette smoking.

CLINICAL BOTTOM LINE

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What measures of pneumonia care are regularly collected and publicly reported?

In the past, the Center for Medicare & Medicaid Services (CMS) and the Joint Commission have endorsed a set of "core measures" for CAP, which focused on vaccination, smoking cessation, antibiotic choice, timing of therapy, and doing blood cultures on critically ill patients before the start of antibiotics. The performance of individual hospitals in achieving these measures had been publicly reported. As of 1 January 2015, data collection on these measures was suspended because compliance was high and continued monitoring and public reporting were not believed to be valuable. The quality measures recommended by professional organizations are being reevaluated, as are the guidelines for the treatment of CAP. The CMS will continue to monitor all-cause mortality rates for patients with CAP, and they will monitor, and tie future reimbursement rates to, 30-day readmission rates for these patients.

In the Clinic Tool Kit

Community-Acquired Pneumonia

Patient Information

www.nlm.nih.gov/medlineplus/ency/article/000145.htm
NIH MedLine Plus.

www.cdc.gov/pneumococcal/clinicians/diagnosis-medical-mgmt.html
Centers for Disease Control and Prevention.

http://familydoctor.org/familydoctor/en/diseases-conditions/pneumonia.html

http://umm.edu/health/medical/ency/articles/pneumonia-adults-community-acquired
Patient resources in English.

http://umm.edu/health/medical/spanishency/articles/neumonia-en-adultos-extrahospitalaria
Patient resources in Spanish.

Guidelines

https://www.thoracic.org/statements/resources/mtpi/idsaats-cap.pdf

https://www.brit-thoracic.org.uk/guidelines-and-quality-standards/community-acquired-pneumonia-in-adults-guideline/
Medical guidelines for clinical practice for the diagnosis and treatment of community-acquired pneumonia.

In the Clinic

WHAT YOU SHOULD KNOW ABOUT COMMUNITY-ACQUIRED PNEUMONIA

What Is Pneumonia?

Pneumonia is a serious infection of the lungs. Community-acquired pneumonia is a type of pneumonia that you develop outside of a hospital or nursing home. It is most commonly caused by bacteria, but viruses can also cause it. Risk factors for pneumonia include:

- Being 65 years or older
- Eating a poor diet
- Drinking alcohol
- Smoking cigarettes
- Having influenza (the "flu")
- Having a weakened immune system, such as from old age or a serious disease, like HIV or cancer
- Having periods of time where you lost consciousness, such as from anesthesia, drinking too much, drug use, stroke, and seizure

What Are the Warning Signs of Pneumonia?

- Long-lasting cough, which can sometimes bring up mucus
- Feeling cold and shaky
- Fever
- Shortness of breath
- Chest pains
- Feeling tired and weak

How Is Pneumonia Diagnosed?

- Your doctor will ask you questions about your symptoms and give you a physical examination. He or she will listen to your lungs and heart and check your temperature.
- Your doctor will usually order a chest X-ray to see how much your lungs are affected.
- Your doctor may also order tests of the sputum (mucous brought up with coughing) and urine to learn what type of bacteria is causing the pneumonia. A blood test can show if the infection has spread from the lungs to the blood.

How Is Pneumonia Treated?

- If your pneumonia is caused by bacteria, your doctor will prescribe antibiotics. Your symptoms usually start to go away within a few days of starting treatment. It is important to finish all of your antibiotics, even if you are feeling better.

- Drink lots of fluids to make sure you stay hydrated.
- Your doctor may prescribe medicines for cough and to reduce fever. Feeling tired and coughing may last for up to a month or longer before going away.
- Most patients are treated at home, but some who are very ill or who have a greater risk for complications may have to stay in the hospital.
- If you have to stay in the hospital, doctors will monitor your heart and breathing rates, oxygen levels, and temperature. You may also be given fluids and medicines through your veins (intravenously).

Questions for My Doctor

- Am I contagious?
- What can I do to help relieve my symptoms?
- Why do I still feel so tired?
- How can I prevent another episode of pneumonia?
- Should I be treated at home or in the hospital?
- When can I start my normal activities again?
- When can I go back to work?

Bottom Line

- Pneumonia is a serious infection of the lungs. It is most commonly caused by bacteria.
- Symptoms include shortness of breath, coughing, chest pains, high fever, and feeling cold and shaky.
- Treatment usually includes antibiotic medicines, medicines for cough and fever, drinking lots of fluids, and rest.
- Some people may need to stay in the hospital if the infection is more serious or if pneumonia does not go away.

For More Information

ACP℠
American College of Physicians
Leading Internal Medicine, Improving Lives

Medline Plus
www.nlm.nih.gov/medlineplus/ency/article/000145.htm
American Lung Association
www.lung.org/lung-disease/pneumonia/symptoms-diagnosisand.html

Patient Information

Appendix Table. **Drug Treatment for Community-Acquired Pneumonia**

Agent	Mechanism of Action	Dosage	Benefits	Side Effects	Notes
Antibiotics for community-acquired MRSA Linezolid (Zyvox)	Bacteriostatic; binds to 50 S ribosomal subunit to inhibit bacterial protein synthesis	600 mg PO or IV q12h	Penetrates well into the lung and is active against MRSA	Myelosuppression—particularly thrombocytopenia, vomiting, diarrhea, seizures, hypoglycemia	Drug interactions may lead to serotonin syndrome. Do not use with tricyclic antidepressants, monoamine oxidase inhibitors, or selective serotonin reuptake inhibitors. Monitor blood pressure
Clindamycin* (Cleocin)	Bacteriostatic; binds to 50 S ribosomal subunit to inhibit bacterial protein synthesis	600 mg PO q8h	Can inhibit toxin production by MRSA	Diarrhea, esophagitis, hypersensitivity	Can cause *Clostridium difficile*–associated diarrhea. Caution with asthma, severe hepatic disease
Vancomycin (Vancocin)	Bactericidal; inhibits cell wall and RNA synthesis	15–20 mg/kg IV q8–12h	Active against MRSA, with extensive clinical experience. Is not optimal therapy for methicillin-susceptible *Staphylococcus aureus*	Ototoxicity, nephrotoxicity, neutropenia, nausea, hypokalemia	Avoid rapid infusion (causes histamine release). Avoid extravasation. Monitor trough concentrations. In chronic kidney disease, individualize dose
Antipseudomonal β-lactams Piperacillin/tazobactam Cefepime Imipenem Meropenem	Bactericidal, interferes with peptidoglycan cross-linking, and prevents formation of the bacterial cell wall	Piperacillin/tazobactam, 3.375 mg q4–6h IV. Cefepime, 1–2 g q12h IV Imipenem, 1 g q8h IV Meropenem, 1 g q8h IV	Active against pneumococci and *Pseudomonas aeruginosa*	Anaphylaxis, rash, nausea, vomiting, diarrhea, phlebitis, seizures (high doses), hypokalemia (high doses), elevated liver tests, prolonged prothrombin time (especially if on coumadin). Seizure potential greater with imipenem than meropenem	Only use for patients with pseudomonal risk factors, although generally active against DRSP. Can dose daily if patient has renal insufficiency
Cephalosporins Cefuroxime Cefpodoxime Ceftriaxone Cefotaxime	Bactericidal, interferes with peptidoglycan cross-linking, and prevents formation of the bacterial cell wall	Cefuroxime, 500 mg bid PO. Cefpodoxime, 400 mg bid PO. Ceftriaxone, 1–2 g q12–24h (usually q24h). Cefotaxime, 1 g q8h	Active against pneumococci and *Haemophilus influenzae*, including β-lactamase-producing organisms	Anaphylaxis, rash, nausea, vomiting, diarrhea, elevated liver function test results, interstitial nephritis, altered coagulation, pseudomembranous colitis	Not to be used alone in CAP. Combine with a macrolide. Although cefuroxime can be used as oral therapy, it should not be used IV, because it is not as active against DRSP as other cephalosporins
Glycylcycline Tigecycline* (Tygacil)	Bacteriostatic; binds to 30 S ribosomal subunit to inhibit bacterial protein synthesis	100 mg IV initially, then 50 mg IV q12 h. Infuse over 30-60 min	Can be used for CAP, but use only when there are no other options. Penetrates well into respiratory secretions; active against pneumococcus and atypical pathogens, but not *P. aeruginosa*	Vomiting, diarrhea, hepatotoxicity, pancreatitis, anemia. Increase in all-cause mortality	Avoid with pregnancy. With severe hepatic disease, use 25 mg for maintenance infusion
Macrolides Azithromycin Clarithromycin	Bacteriostatic, binds to the 50S ribosomal subunit, and inhibits bacterial protein synthesis	Azithromycin, 500 mg on day 1 (IV or PO), followed by 500 mg (IV or PO) for 7–10 days for hospitalized patients. 250 mg on days 2–5 for outpatients. Azithromycin in the microspheres oral extended-release formulation, 2 g on day 1 (PO) without follow-up dosing for outpatients. Clarithromycin, 500 mg bid PO, or 1000 mg/d PO (extended-release preparation) for outpatients	Covers pneumococcus, atypical pathogens, and *H. influenzae*	Nausea, vomiting, diarrhea, QT prolongation, dyspepsia (clarithromycin)	Use as monotherapy only in patients without cardiopulmonary disease or modifying factors. Otherwise, combine with a β-lactam agent. Erythromycin is less expensive but not recommended because of the need for more frequent dosing, more intestinal upset, and no coverage of *H. influenzae*
Penicillins Amoxicillin/clavulanate Ampicillin Ampicillin/sulbactam	Bactericidal, interferes with peptidoglycan cross-linking, and prevents formation of the bacterial cell wall	Amoxicillin/clavulanate, 875 mg bid PO. Ampicillin, 500–1000 mg tid PO. Ampicillin/sulbactam, 1–2 g q6h IV	Active against pneumococci and β-lactamase-producing *H. influenzae*. High doses (1 g tid) active against DRSP	Anaphylaxis, rash, nausea, vomiting, diarrhea, phlebitis, seizures (high doses), hypokalemia (high doses), elevated liver tests, prolonged prothrombin time (especially if on coumadin)	Do not use alone in CAP. Combine with a macrolide
Quinolones Ciprofloxacin Levofloxacin Moxifloxacin	Bactericidal, interferes with bacterial DNA gyrase. Kills bacteria in a concentration-dependent fashion	Ciprofloxacin, 400 mg q8h IV. Gemifloxacin, 320 mg/d (PO only). Levofloxacin, 500-750 mg/d (IV or PO). Moxifloxacin, 400 mg/d (IV or PO)	Active against *P. aeruginosa*, atypicals and *H. influenzae*. Levofloxacin and moxifloxacin are the "respiratory quinolones" with activity against DRSP, *H. influenzae*, and atypical pathogens	Seizures, hypersensitivity, photosensitivity, tendon rupture, nausea, vomiting, diarrhea, QT prolongation	Ciprofloxacin: Only use in severe CAP. Not always reliable against pneumococci, and should be combined with other agents if DRSP is possible. If used in severe CAP, do not use as monotherapy
Tetracyclines: Doxycycline	Bacteriostatic, binds to 30 S ribosomal subunit, and interferes with bacterial protein synthesis	100 mg bid (IV or PO)	Active against key bacterial and atypical pathogens	Nausea, vomiting, diarrhea, photosensitivity	Not always fully reliable against pneumococci

bid = twice daily; CAP = community-acquired pneumonia; DNA = deoxyribonucleic acid; DRSP = drug-resistant Streptococcus pneumonia; *FDA = U.S. Food and Drug Administration; IV = intravenous; MIC = minimum inhibitory concentration; MRSA = methicillin-resistant* Staphylococcus aureus; *PO = oral; tid = three times daily.*
** Black box warning.*

